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DEPARTMENT OF COMPUTER ENGINEERING

ACADEMIC INTERNSHIP REPORT

STUDY AND IMPLEMENTATION OF STARTECH INTERNS MANAGEMENT SYSTEM

*An internship carried out from the 4th of JULY to the 26th of AUGUST 2022
at STARTECH in the Fulfilment for the award of the Higher National Diploma*

OPTION: SOFTWARE ENGINEERING

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CERTIFICATION

This is to certify that an internship with the theme “**STARTECH**” was carried out from the 4th of July to the 26th of august 2022 and this report was written by **NJUA CYRAN NKWAIN** in partial fulfillment of the requirement for the award of the Higher National Diploma (HND) in Software Engineering.

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MR NATHE Samuel

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PROFESSIONAL SUPERVISOR:

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DATE

DEDICATION



TO MY LOVELY
FAMILY

ACKNOWLEDGMENT

The accomplishment of this work required the assistances of some people. Failing to quote them individually here, they receive my most sincere thanks. I am particularly grateful to:

- Mr. CEDRIC DJAMBOU, President of IUG for the establishment of this structure for the purpose of training students.
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- All my FRIENDS and CLASSMATES who contributed in one way or the other to the realization of my project.

ABSTRACT

The management of interns is important for every company; it can decide the success of the training the company gives to its interns and whether they really acquire the knowledge or not. A company will have to ensure a good management of its intern's information and their attendance in order to maximize the quality of training they offer. Also in Managing interns 'information, their presence and even the payments of fee have become a great challenge because many companies do it manually with a high risk of errors. For the reasons indicated, the company STARTECH offers us the permission to create an effective web application that will help manage all its interns information. This application registers both interns and trainers record attendance of each training day electronically and stores it in a database. The system is designed using Model, View and Template (MVT) architecture and implemented using the power of Python and Django. JavaScript is added to the application to improve system usability. Sqlite3 is used for the application database. Different presentations of information can be obtained from the system. The system test cases revealed that the system works tremendously and is ready to be used by STARTECH academy to manage its interns. For further research, I plan to add an online payment method so that people ready to obtain an internship at StarTech Academy can directly pay for it online.

RESUME

La gestion des stagiaires est importante pour chaque entreprise ; il peut décider du succès de la formation que l'entreprise donne à ses stagiaires et s'il réussit ou non à acquérir réellement les connaissances. Une entreprise devra assurer une bonne gestion des informations de ses stagiaires et de leur assiduité afin de maximiser la qualité des formations qu'ils proposent. Aussi dans. Gérer les informations des stagiaires, leur présence et même le paiement des honoraires est devenu un grand défi car de nombreuses entreprises le font manuellement avec un risque élevé d'erreurs. Pour les raisons indiquées, la société STARTECH nous offre l'autorisation de créer une application web efficace qui permettra de gérer toutes les informations internes. Cette application enregistre électroniquement la participation des stagiaires et des formateurs à chaque journée de formation et la stocke dans une base de données. Le système est conçu à l'aide de l'architecture Model View and Template (MVT) et de la mise en œuvre à l'aide de la puissance de Python et de Django. JavaScript est ajouté à l'application pour améliorer la convivialité du système. SQLite3 est utilisé pour la base de données de l'application. Différentes présentations d'informations peuvent être obtenues à partir du système. Les cas de test du système ont révélé que le système fonctionne énormément et est prêt à être utilisé par l'académie STARTECH pour gérer ses stagiaires. Pour des recherches plus approfondies, je prévois d'ajouter un mode de paiement en ligne afin que les personnes prêtes à obtenir un stage à la StarTech Academy puissent directement le payer en ligne.

PREFACE

Founded in 1993, the University Institute of the Gulf of Guinea is a group of three higher education institutions based on the same campus in Basa-Douala. Specialized in fields as varied as Commerce and management, communication and information, industry and new technologies, and paramedical training, its offer allows students to adapt their course as they evolve.

IUG trains young professionals in the Technical, Medical and Business environment. Since its creation in 1993, the advance school of management has been striving to answer the urgent calls of economic operators in need of foremen and management staff. With the launching of the HIGHER NATIONAL DIPLOMA, the university Institute of the Gulf of Guinea is becoming the first bilingual Higher Institution of learning in central Africa.

The Cameroon government has taken as duty to train and educate youths in all fields of vocational training, in order to build up their professional skills. IT IS IN THIS LIGHT THE ORDER NO 008/CAB/PR of 19/03/1993, brought about the creation of private state university and higher professional institute among which is ISTA (INSTITUT SUPÉRIEUR DESTECHNOLOGIES AVANCÉES). ISTA was created by the ministerial order No 05/0038 of the 12th January 2005. It had as mission to employ good lecturers who are devoted and willing to train students to obtain the higher national diploma (HND), after a two-year course. Below are some of the professional trainings offered in IUG.

Computer Engineering, Accountancy, Banking and finance, Transport and logistics, Electrical Power System, Telecommunication, Nursing.

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LIST OF ABBREVIATIONS

IUG: University Institute of The Gulf of Guinea

ISTA: Higher Institute Of Advanced Technologies

JS: JavaScript

HTML: Hypertext Markup Language

CSS: Cascading Style Sheet

AMD: Advanced Micro Devices

DB : Database

PDF : Portable Document Format

TXT: Text

XML: Extensible Markup Language

SDK: Software Development Kit

CPU: Central Processing Unit

SQL: Structured query Language

AJAX: Asynchronous JavaScript and XML

DBMS: Database Management System

IDE: Integrated Development Environment

SSD: Solid State Drive

DDR4: Double Data Rate 4

SDLC: System Development Life Cycle

UML: Unified Modelling Language

GUI: Graphical User Interface

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GENERAL INTRODUCTION

This report is a short description of our two-months internship carried out as compulsory component of the HND examination. The internship was carried out at STARTECH from the 4th July to the 26th AUGUST 2022.

Interns' management system is a software developed to manage the daily activities of interns in STARTECH academy. It facilitates in accessing personal information of a particular intern and attendance of a particular day in a particular year. The information is sorted by the operators, which will be provided by the administrator in charge of the daily management of interns. This system will also help in generating a report concerning a particular intern in the academy. The main purpose for this academic internship was for us to learn how to apply our academic skills acquired to the professional world. In addition to that, we had to learn new things that was very important for us to know and was not taught in school. We will be introducing the project and research. We will further look at the necessity of this study, research objectives, questions and significance. Last but not the least, this report gives the detailed management of every intern that does internship in STARTECH academy.

THE PROBLEM STATEMENT

Managing interns in a company seems to be an easy activity but while requires lot of care to be taken. Like registering personal information of an intern it's obvious that there can be mistakes upon the registration, so there should exist an easy way to solve the problem or remodify the information to the correct ones. Also interns management also involves keeping track of the records of those who completed their fee or keep track of those still owing an amount to complete and the amount they are owing, and it also involves keeping track of attendance of the interns in the academy so that a good judgment of weather an intern really acquire the skills or not can be done so as to generate a good report concerning the intern.

Many companies not only use the attendance to generate report but also to create a reward system for younger interns so as to emphasize the importance of consistent attendance. Because many intern may be missing training sessions for kind of reasons, from illness to family obligations so keeping track of attendance can make the trainers to take notice of issues sooner and be able to work with interns to address them.

We can see that managing interns' information, fee, and attendance of a company with different fields of training and each field containing more than 30 interns is not really an easy task to do and that is where our problem comes. The problem is how we can manage the information of many interns and attendance of interns in a particular field.

AIM OF THE STUDY AND OBJECTIVES

The main purpose of developing an intern management system is to computerize the tradition way of keeping interns' information and another purpose of developing this system is directly and easily generate a report concerning an intern in the company

The objective to create an intern management system is to perform the following tasks:

- To register both interns and trainers in a particular field in the company
- To display both the list of interns and trainers of every specialty in the company
- To facilitate the records of fee payment
- To enable retrieval of information when ever needed from the system.
- To also reduce human error
- To maintain a digitalize record of daily attendance.

RESEARCH QUESTIONS:

- How can we free information from human error making?
- How can we avoid manual or traditional recording of attendance?
- How can we improve the management of interns' information?
- How can we avoid time consumption made during data recording?

SIGNIFICANCE OF STUDY:

1. The research will enable the board of directors to be aware of the problem of manual record keeping and any improvement made by computerized record keeping in order to find appropriate steps in addressing them.
2. To understand the importance of a good management of interns' information in a company.
3. It will help other companies to also adopt and implement these recommendations to improve their systems.

STRUCTURE OF THE WORK IN PARTS

This work has two parts with two chapters each, details are as follows:

- ✓ Chapter one is the general presentation of the enterprise that is; Historical background, products and services, organizational structure, functions of departments and geographical location.
- ✓ Chapter two will present an introduction to the literature review, the theoretical review and internship activities.
- ✓ Chapter three deals with the presentation of methodology of research. That is a recall of the objectives, resources of data, definition of terms, methods of data collection, methods of data analysis and the result obtained.
- ✓ Chapter four finally will give a summary of the problems encountered during the research, recommendations to and a conclusion of the research.

FIRST PART
THE FRAMEWORK (ENVIRONMENT) OF
THE TRAINING COURSE

CHAPTER ONE

GENERAL PRESENTATION OF THE COMPANY

Chapter one is made up of two sections, section 1 and section 2. Section 1 talks about how STARTECH was created as well as its organization, and its basic aspects and section two talks about the external environment which include competitors, suppliers.

The StarTech academy is a limited liability company specializing in the provision of services and training in the field of ICT. The company has been operating in Cameroon for almost four years already and is registered in the trade register with the number: **RC / DLA / 2021 / B / 3445**.

SECTION ONE: CREATION AND EVOLUTION OF THE COMPANY

1.1.1 CREATION OF THE COMPANY

Following a clear observation which is that of the glaring unemployment of young graduates, the company has decided to contribute in its own way to reduce the phenomenon which is constantly growing. In fact, we realized that the majority of students go to school with the sole objective of obtaining their diploma at the end of the year. This and these do not give any importance with regard to the acquisition of practical know-how. This is why we find more and more in our environment people who claim to be engineers and yet they are unable to produce scientific work up to the skills they are supposed to have. This is why, as a corporate citizen, the StarTech Academy's mission is

to support these young learners with a view to initially enabling them to acquire a practical skill thanks to the training sessions that we provide. organize and then integrate our various projects in progress.

1.1.2 EVOLUTION OF THE COMPANY

The StarTech Academy since its creation until today has stood out thanks to a number of achievements among which we can mention among others:

- School management software: Phoenix Shoomann functional in 05 schools only in the city of Douala.
- Commercial management software: LOGESCO functional in two shops in the city of Douala.
- Pharmacy management software: PHARMA-SOFT functional in pharmacies in Yaoundé and Ebolowa.
- A tontine management software: functional LOGIT in a meeting group in the quarry district in Douala.
- A mobile application for the sale and exchange of articles: BiZi available on the stores (APP store and Play store).
- A mobile application for learning to cook: Gooo available on the stores (APP store and Play store).

Besides computer programming, we also excel in network security and administration, computer maintenance, home automation and even renewable energies. In a very short time, we expect to really establish ourselves in the software market with the creation of a whole specific department for the development of “Made in Cameroon” applications.

SECTION 2: ORGANIZATION AND OPERATION

1.2.1 ORGANIZATION

The organizational structure of StarTech Academy includes five (5) main departments namely:

- **StarTech Software**
- **StarTech Network**
- **StarTech Computer**
- **StarTech Energy**
- **StarTech Academy (General Manager)**

➤ **StarTech Software**

The “StarTech Software” department is responsible for producing applications for any type of environment. The applications in question can be:

- **Mobiles:** These are applications running on Android & IOS terminals.
- **Web:** Web applications are those running through a browser by means of web pages. Websites also fall into this category.
- **Desktop:** Desktop applications are those that can be installed on a computer.
- **Home automation:** Home automation applications make it possible to control the electronic systems that we design.

This department is also responsible for designing and administering databases (SQL, NoSQL).

➤ **StarTech Network**

The main role of this department is to design the network architecture for companies. These main missions among others are:

- Design custom network architectures for companies.
- Perform the installation and maintenance of corporate networks.
- Implementation of network services (Telephony, Mail, Internet, etc.).
- Network administration (Windows, Linux, Cisco...).
- Installation of video surveillance and access control systems.
- Audit of security systems.

➤ **StarTech Computer**

Through this department, we offer maintenance services for computer equipment. Our services are:

- Hardware maintenance: troubleshooting, assembly and disassembly of computer equipment.
- Software maintenance: Installation and updating of operating systems, management of computer parks, etc.

➤ **StarTech Energy**

Revolutionary department which mainly works in the field of renewable energies. The main missions are:

- Electrical network installation thanks to solar panels.
- Maintenance and supervision of the electrical network.

➤ **StarTech Academy**

The academy responsible for carrying out training is managed by the general management of the company. The academy's missions are:

- Offer practical and innovative training in the fields of ICT
- Identify the best learners and integrate them into our various projects.
- Support project leaders who need our expertise.

1.2.2 FUNCTIONING

The company regularly organizes training sessions and practical workshops with the aim of helping all people (students or not) to acquire real knowledge and skills. We also offer vacation internships for students to allow those to experience the realities of the business world. It should be noted that at the end of our various training sessions we integrate the best students into our various projects in progress.

The StarTech Academy with the aim of always satisfying customers, works daily around the objectives and values that it has set for itself.

➤ **Goals and values**

• **Goals**

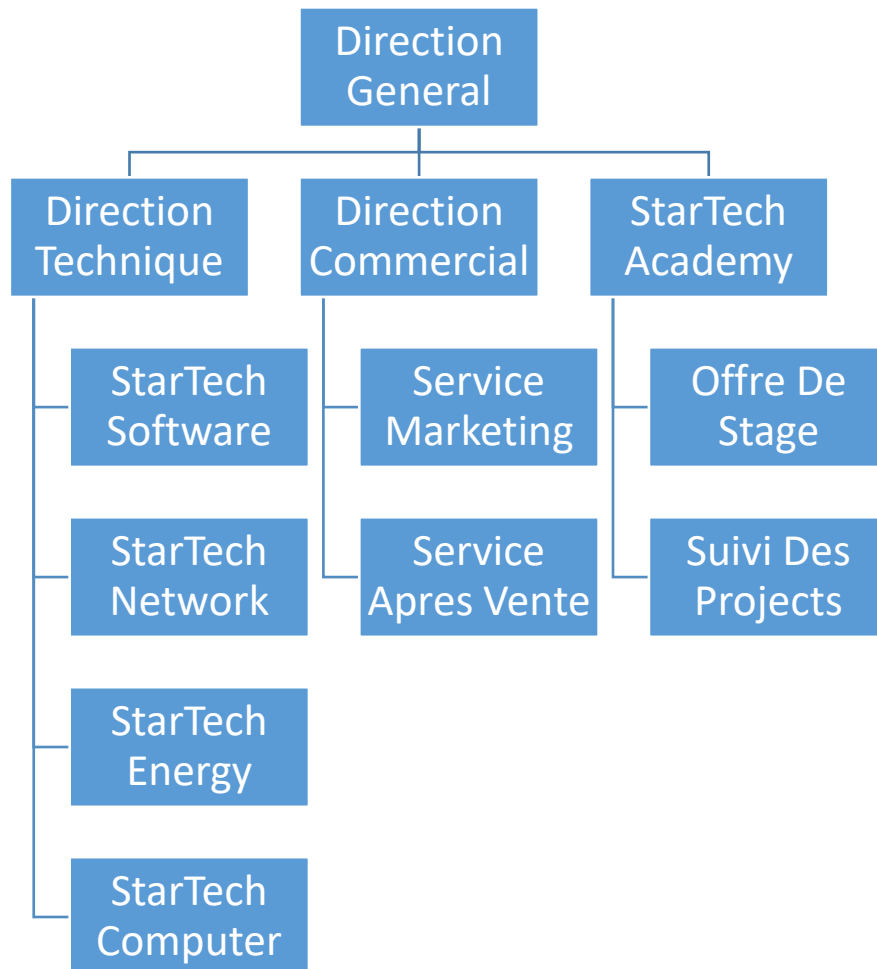
Training-Research-Innovation is our motto and to be able to honour it, the company has set itself the following objectives:

- ✓ **Satisfy customers in general businesses by offering a quality service respecting International standard.**

- ✓ **Respect of delivery deadlines accompanied by an efficient after-sales service to ensure customer satisfaction**
- ✓ **Provide innovative solutions to the ills that plague our society.**
- ✓ **Offer quality training to our different learners.**
- ✓ **Continuously monitor technology.**
- **Values**
 - ✓ **Quality:** The StarTech Academy places a lot of emphasis on product quality and service efficiency. This quality of products and services requires respect for procedures, but also the quality of the people: you, your skills and your creativity.
 - ✓ **Creativity:** StarTech Academy considers the creative act to be the very essence of its existence. Thanks to your enthusiasm, your inspiration and your deep emotions, the company believes you are capable of transforming information into real development strategies.
 - ✓ **Team spirit:** thanks to participatory management and thanks to the team spirit, the company wants everyone to contribute to achieving their vision. It is the effort of each at his level that will allow the development of the company.
 - ✓ **Integrity:** this is one of the essential human qualities in the professional life of StarTech Academy. More than a simple image, integrity brings the confidence of peers and the hierarchy, but also it gives the company a solid reputation.

➤ **Organizational chart**

Figures 1: Organization chart of StarTech Academy



- **General management:** It develops general managerial policy, defines short, medium and long-term objectives and coordinates all of the company's activities.

- **Technical direction:** the technical direction is at the centre of the operation of the company. This is where all projects are thought out, planned and carried out. it contains all the main departments of the company and obviously works under the watchful eye of senior management.
- **Sales Department:** It defines the sales and marketing policy for products and services, takes care of the permanent search for new customers and also ensures the effectiveness of the after-sales service.
- **Academy StarTech:** its mission is to offer innovative training in different fields of N.

CHAPTER TWO

COURSE OF THE INTERNSHIP

This chapter is made up of two sections. Section 1 which gives a Brief summary on the reception and Section 2 which gives a Brief summary of the internship activities carried out by the intern during his internship and justification of topic studied.

SECTION 1: PRESENTATION OF SERVICE OF RECEPTION

On the first day of internship, we were pleasantly received by the members of the staff and they further introduced themselves by stating their names, years of experience in the field, projects done and projects still in completion. Also, we were asked to introduce our self by stating our names, school, years of experience in the field, projects completed (if any) or projects still in process. Again, we were asked the reason for choosing to do internship here and the objectives to accomplish after doing internship there.

Furthermore, after the general introduction of both the interns and the members of the staff, we later met with our respective professional supervisor. We were given an oral quiz to evaluate and test our competency. A structure was laid out on how the program would look like throughout the internship period.

SECTION 2: ACTIVITIES OF THE TRAINING COURSE

This section is further divided into two parts which are: Chronological program of activities and effectiveness and the justification of the topic.

Part 1: Chronological program of activities and effectiveness

This section provides the activities performed by the intern during his two months of internship. This report is based mostly on the activities that were carried out during my internship period which ran from July to August 2022 at StarTech Academy.

Period	Activities	Results	Difficulties Encountered	Solutions
WEEK 1 4-07-2022 To 9-07-2022	Introduction and presentation of interns and staff members.	The presentation when well and each one presented him or herself.		
	Notion of HTML, CSS and JavaScript.	Notions of HTML, CSS and JavaScript was clearly understood.	Working with functions in JavaScript	Research on the topic
	Project 1: Creation of a webpage with Html, CSS	We took less than two days to finish the webpage with notions acquired.	Time management	Design a study timetable
	Correction of project 1 and additional sub projects.	The correction when perfect but some had some difficulties with JavaScript.		
WEEK 2 11-07-2022 To 16-07-2022	Project 2: Creation of the Facebook homepage with HTML, CSS and JavaScript from Scratch.	Project 2 was realized in 2 days and each one had something to present.		

	Project 3: Recreation of Facebook home and login page with the Notion of framework like Bootstrap 4 and Latest Version. (Bootstrap5)	The projects were not realized by all the member or the interns which were doing the same specialty with me, but I did succeed.	Was new in using library of code like that of Bootstrap and was not easy to manipulate class name	Had to revise the Bootstrap documentation. And work with more tutorials
	Project 4: Applying the Bootstrap and JavaScript on a webpage of our choice.	Presentation of each member of the group successfully took place and both the directors and prof. supervisors of each intern came.		
WEEK 3 18-07-2022 To 23-07-2022	Project 5: Creation of Administrator Dashboard with his various functions of different components.	Each and every one presented his or her design admin dashboard to the supervisor.	Some interns find it difficult to design the dashboard	Ask for help from friends who succeeded well in the design

	Project 6: Creation of Registration form with link to dashboard with HTML CSS and JavaScript and Bootstrap 4.0	Each and every one presented his or her design admin dashboard to the supervisor.		
WEEK 4 25-07-2022 To 30-07-2022	Introduction to back-end Web Development with PHP and MySQL.	Since it was new, we had difficulties in understanding it the first day but when smooth the following days.	Difficulties in understanding the form MVC	Research made
	With the use of the notion of PHP and MySQL, creation of a back-end with all the project done on Week3, project 6(registration form and dashboard)	All our project made from Week 1 to Week 4 was updated by adding a backend code to it even if it was just to print out an output.		
	Project 7: User Input verification and Validation phase	The project when all well and each and every one had an idea on how to implement the verification and validation phase during form submission.		

WEEK 5 & WEEK 6 1-08-2022 To 13-08-2022	Introduction to our various HND project	Supervisor propose some projects for those who were in need and concerning those whom already had an HND project, they detailly explained the project to him.		
	Requirement Analysis and Feasibility Study	The task was successfully complete by each intern.		
	System Analysis and System design	Basic knowledge was acquired but still was theoretically.		
	Worked on login authentication.	We had a good understanding of how to authenticate our login pages		

WEEK 7 15-08-2021 To 20-08-2021	Beginning of implementation of our HND projects based on the knowledge we acquired so far.	Every intern started implementing their projects with the necessary knowledge acquired during the ongoing of the internship.		
	Presentation of the first page of our project created.	Every intern presented a home page of their project		
WEEK 8 22-08-2022 To 26-08-2022	End of PHP	We had good knowledge on how to submit a form, interact with the relational database by performing the CRUD operations.		
	Continuation of work on our HND projects	With the additional Knowledge on crud operation we were able to continue adding more functionalities on our projects		
	Second presentation of the level attend on our different projects	Every intern improved on their work on project		

	Delivery of interns' attestations	Every intern received an attestation of internship and in addition to that, the owner of StarTech Academy organized a little party for us		
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Table 1: Internship activities carried out in the company.

Part 2: justification of the internship report topic (mission)

An intern management system provides every information about an intern and the daily attendance of all the interns in the company. It is efficient in the fact that less resources and energy is needed to record all these information as compared to the manual way in which it was done years back. Now with this system it's much easy to register an intern in the company and his information can be clearly seen with just a click and can even easily generate a report about the intern he wish just by going to the system tape the name of the intern he wish to generate the report and add some little information like his grade supervisor name and year to the one's that exist already and print

Furthermore, a request for an internship can be done online by just going to the contact page of the system and sending an email to the admin for the request . This is also valid for a trainer who wish to train interns in the company. In addition, it reduces human error that is some unintentional error human can make when either recording the attendance of a day in a particular specialty and also when registering an intern. To conclude, this system facilitates live by changing all what

was done manually to digitalize form there by developing our country as technology innovates.

facilitates live by changing all what was done manually to digitalize form there by developing our country as technology innovates.

SECOND PART: PRACTICALS AND FIELDWORK

CHAPTER THREE

PRACTICAL PHASE

This chapter is made up of two sections. Section 1 which gives a Brief summary on the project management of the intern's management system including UML diagrams and Section 2 which gives the result obtained.

SECTION ONE: PROJECT MANAGEMENT

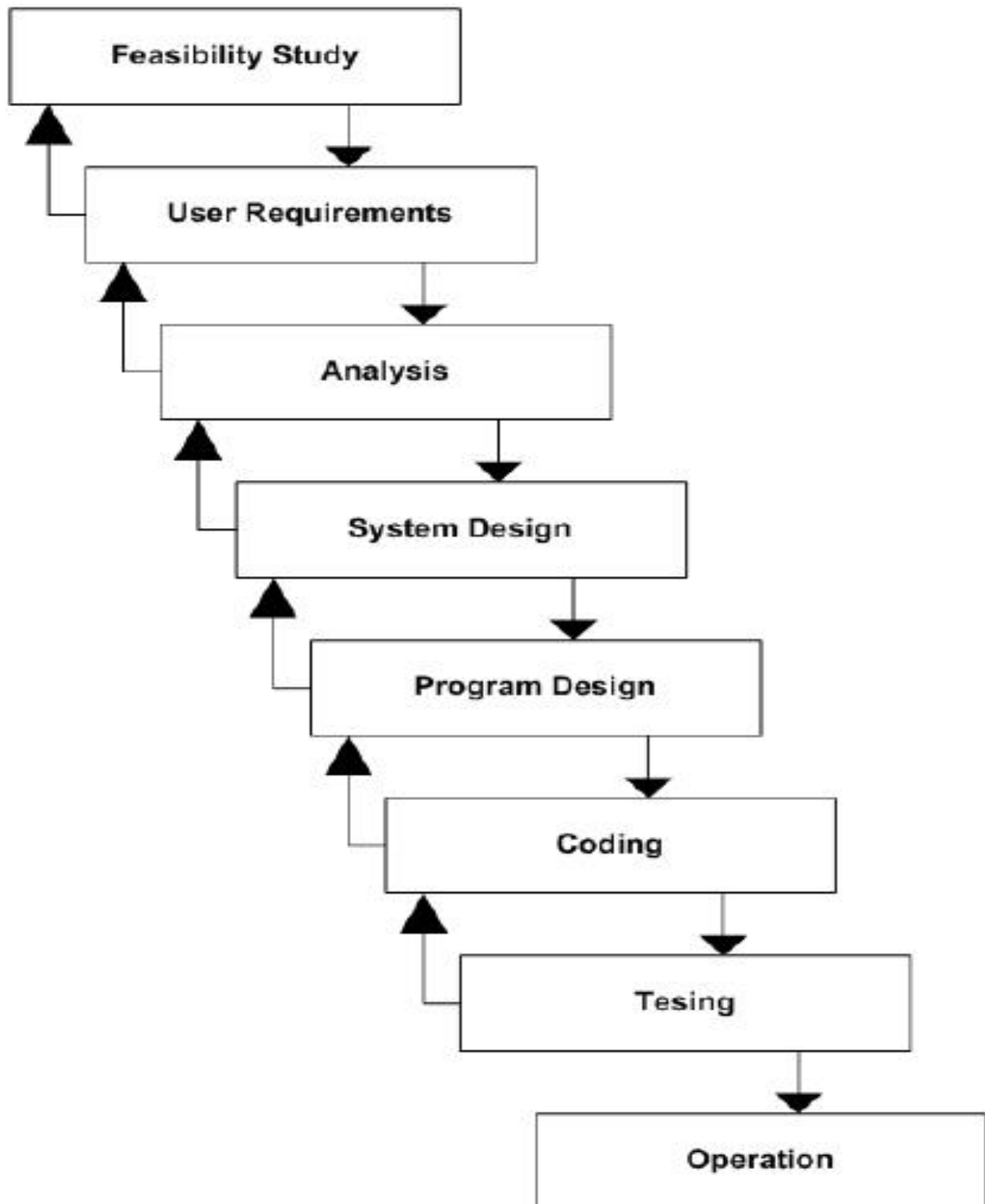
3.1.1 Project development approach

3.1.1.1 *Software Process Model*

To solve actual problems in an industry, software developer or a team of developers must incorporate a development strategy that encompasses the process, methods and tools layers and generic phases. This strategy is often referred to as process model or a software developing paradigm. A process model for software developing is chosen based on the nature of project and application, the methods and tools to be used, and the controls and deliverables that are required. All software development can be characterized as a problem-solving loop in which four distinct stages are encountered: Status, Problem definition, technical development and solution integration. Regardless of the process model that is chosen for a software project all of the stages coexist simultaneously at some level of detail.

This project follows the *waterfall model* which is made up of the following steps: Requirement definition, System analysis, System design, Implementation, Integration and System Testing, Operation and maintenance.

Figures 2: Waterfall Model



Source: Student interns

3.1.2 SOFTWARE DEVELOPMENT LIFE CYCLE MODEL OF THE INTERNS MANAGEMENT SYSTEM (Waterfall Model)

3.1.2.1 Requirement Analysis and Feasibility study

User Characteristics

This system will be use by two users which are the super-admin and Administrators. As all of these have different requirements the modules are designed to meet their needs and avoid any type of confusion. The Uses of all the two User Modules have been described below.

System users can perform the following functions in the system:

- **The Super Administrator**
 - Can login into the system.
 - Can view and modify his profile.
 - Can carry out CRUD on administrator and specialty.
 - Can view both the list of all interns and Trainers.
 - And he can also logout.
- **The Administrator**
 - Can login into the system.
 - Can view and modify his/her profile.
 - CRUD on interns, trainers, and specialties.
 - Generate a printable report of an intern.
 - Make daily attendance for all specialty and generate report.
 - Can add the content of the about page.
 - And can logout.

System Requirement

1. Hardware:

- a. Pentium IV
- b. 1GB RAM
- c. 2GB hard drive space

2. Software:

- a. Operating system: Windows 7, 8, 10, 11 or later version or MacOS X.
- b. Front-end: HTML, CSS, JavaScript, Bootstrap 4 and 5.
- c. Back-end: Python, Django and Sqlite3 as database

External Requirements:

i. Ethical Requirements:

Product Software should be negotiated with human Ethics, and it is wished that the software should be used with Ethical requirements. Any non-ethical use of this software will be at the user's own risk.

ii. Legislative Requirements:

➤ Privacy Requirements:

Privacy of any particular things and strategy will be served based on prior negotiation and it should be cleared purposed for that requirement.

➤ Safety Requirements:

Safety will be provided based on the requirement of the members and necessary critical points.

3.1.2.2 System Analysis

• STUDY AND WEAKNESS OF THE CURRENT SYSTEM

➤ Current System

In the present system all work is done on papers. All interns and trainers' information are stored in a register and the whole attendance of all specialties has also just been recorded in an attendance register and at the end of the internship another work is done for transferring back the information's into The computer to generate interns report which is time and energy consuming.

➤ Weakness of the current system

1. Not User Friendly:

The existing system is not user friendly because the retrieval of data is very slow, and data is not maintained efficiently.

2. Difficulty in report generating:

All calculations to generate report are done manually so there is greater chance of errors.

3. Lots of paperwork:

Existing system requires lot of paperwork. Loss of even a single register/record led to difficult situation because all the papers are needed to generate the reports.

4. Time consuming:

Every work is done manually which takes a lot of time during the calculations of the daily attendance of each intern of each specialty to generate their report.

➤ **REQUIREMENTS OF THE NEW SYSTEM**

1. User Requirement

The User requirements for the new system are to make the system fast, flexible, less prone to errors and reduce expenses and save time.

- Time can be saved in the cost of registering an intern or a trainer and during the printing of his report.
- A system that will reduce all the expenses that are been incurred in the cost of buying papers or registers for registration of interns and trainers in the company.
- A system that will automatically calculates the fee that is still owed by each intern to the company.
- The New system should be more secured so as to prevent the loss of information's concerning each intern and trainer in each specialty.
- Finally, it should prove cost effective as compared to the current system.

2. FEASIBILITY STUDY

A key part of the preliminary investigation that reviews anticipated costs and benefits and recommends a course of action based on operational, technical, economic, and time factors. The purpose of the study is to determine if the systems request should proceed further.

➤ **Economically Feasibility:**

The system being developed is economic with respect to Business or point of view. It is cost effective in the sense that has eliminated the salary work completely. The system is also time effective because the calculations are automated which are made at the end of the month or as per the employee requirement. The result obtained contains minimum errors and are highly accurate as the data is required.

➤ **Technical feasibility:**

The technical requirement for the system is economic and it does not use any other additional Hardware and software.

➤ **Behavioral Feasibility:**

The system working is quite easy to use and learn due to its simple but attractive interface. user requires no special training for operating the system.

d. Features of the new system

The new system has been designed as per the user requirements so as to fulfill almost all of them.

- **User Friendly:**

The proposed system is user friendly because the retrieval and storing of data is fast and data is maintained efficiently. Moreover, the graphical user interface is provided in the proposed system, which provides user to deal with the system very easily.

- **Reports are easily generated:**

Reports can be easily generated in the proposed system so user can generate the report as per the requirement (end of internship) or in the middle of the session to take some measures concerning that not serious. User can give the notice to the parent of the intern.

- **Very less paperwork:**

The proposed system requires very less paperwork. All the data is feted into the computer immediately and reports can be generated through computers. Moreover, work becomes very easy because there is no need to keep data on papers.

- **Computer operator control:**

Computer operator control will be there so no chance of errors. Moreover, storing and retrieving of information is easy. So, work can be done speedily and in time.

3.1.2.3 SYSTEM DESIGN

Using UML Over MERISE for the development of the system

i. What is UML?

According to Booch (2003), **UML (Unified Modeling Language)** is a general-purpose Visual language that is used to specify, visualize, construct, and document the artifacts of a software system.

It allows the modeling of all the phenomena of the business activity (business process, information system, computer system, software

components, etc..) independently of the implemented techniques. A model is an abstraction of reality, it allows:

- To facilitate the understanding of the studied system (that is to say, it reduces the complexity of the system studied);
- To simulate the system studied. As a result, as part of the modeling of the information system, we will have to determine which approach to use for our modeling.

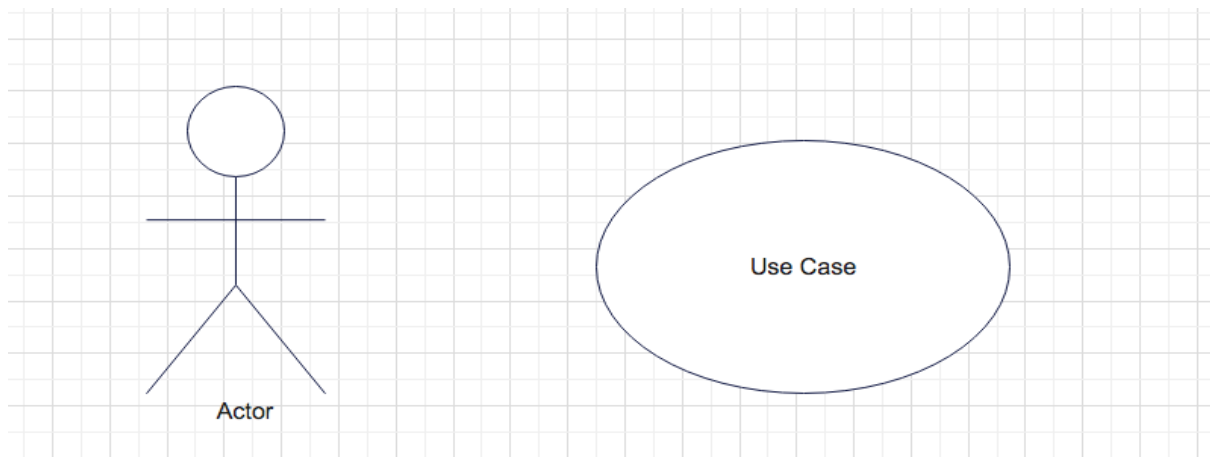
3. UML DIAGRAMS

i. Use Case Diagram

Actors and use cases.

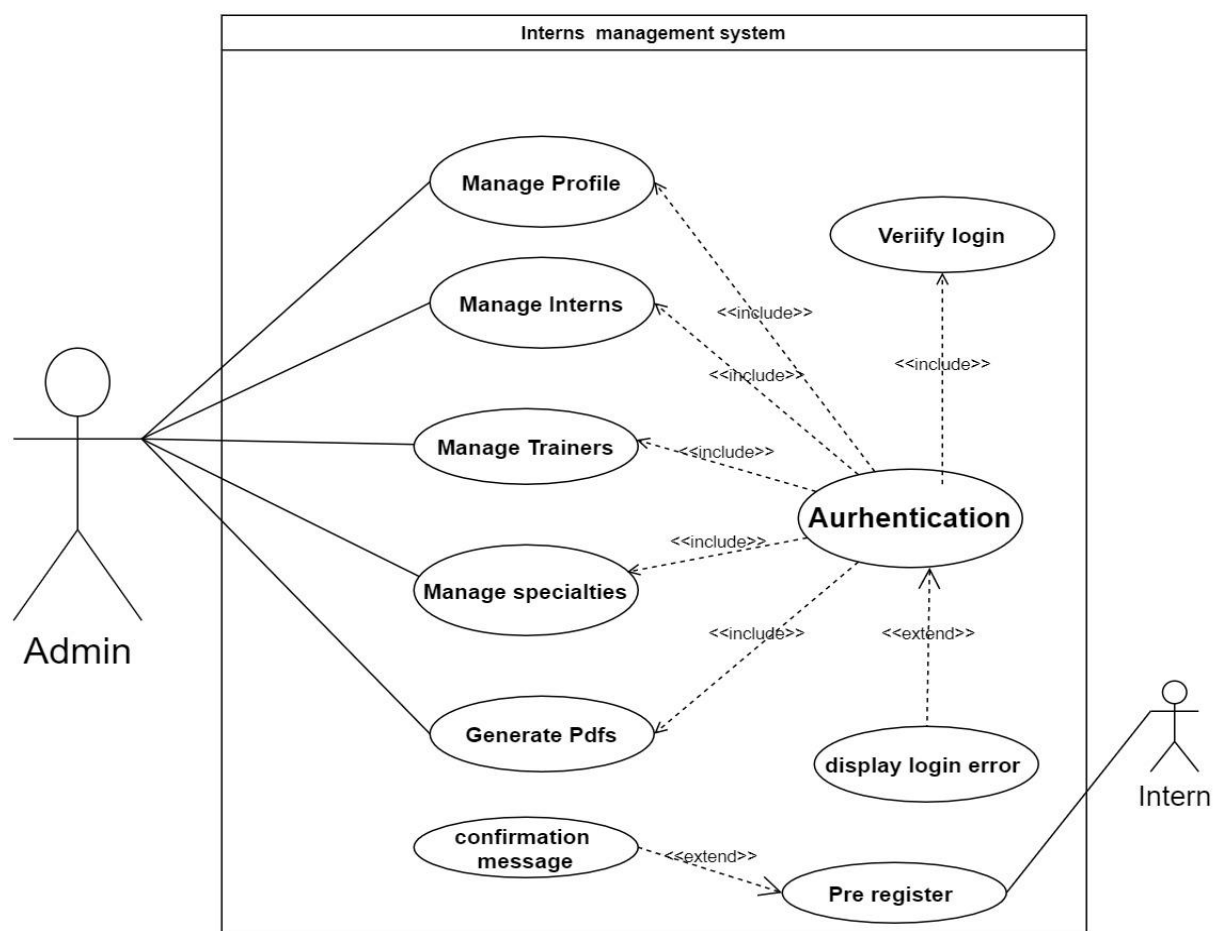
Actors are external entities that interact with the system. Examples of actors include a user role (e.g., a system administrator, a bank customer, a bank teller) or another system (e.g., a central database, a fabrication line). Actors have unique names and descriptions.

Use cases describe the behavior of the system as seen from an actor's point of view. Behavior described by use cases is also called **external behavior**. A use Case describes a function provided by the system as a set of events that yields a visible result for the actors. Actors initiate a use case to access system functionality. The use case can then initiate other use cases and gather more information from the actors. When actors and use cases exchange information, they are said to **communicate**.



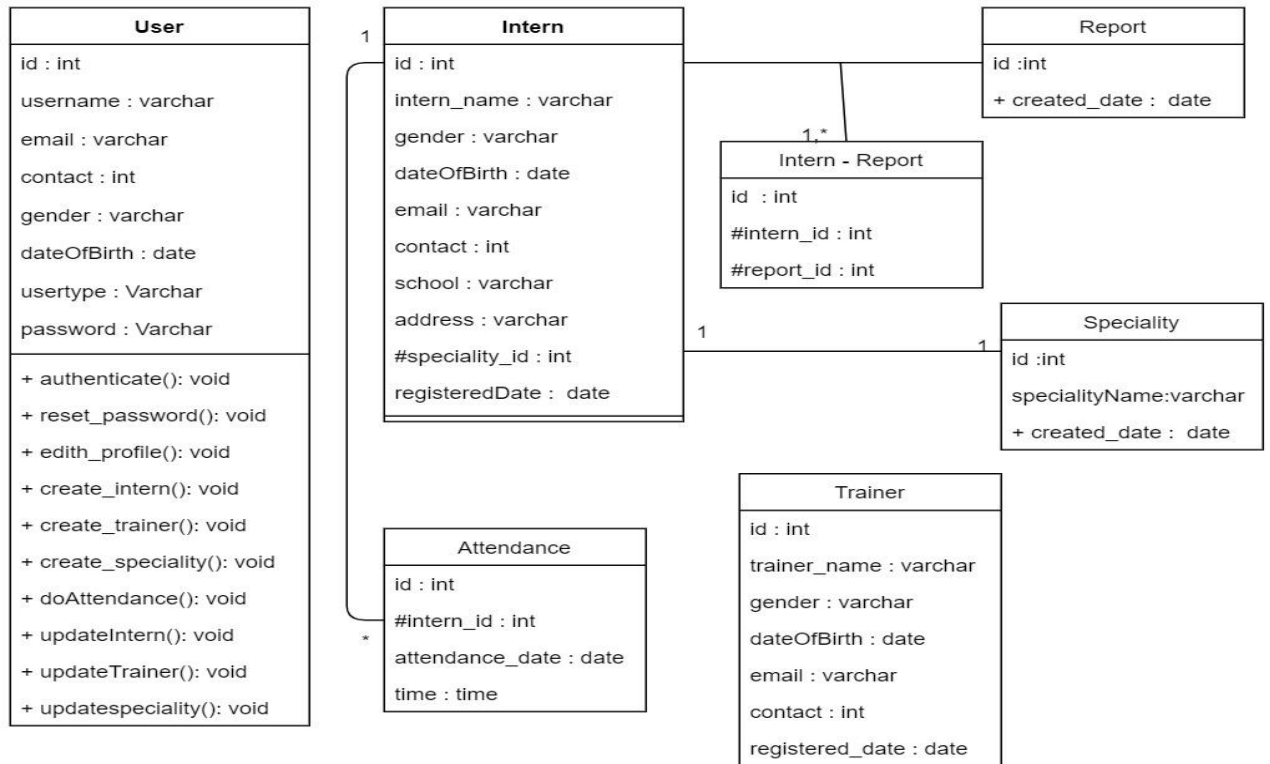
Source: Student interns

Figures 3 : Use case diagram of the super administrator of the system



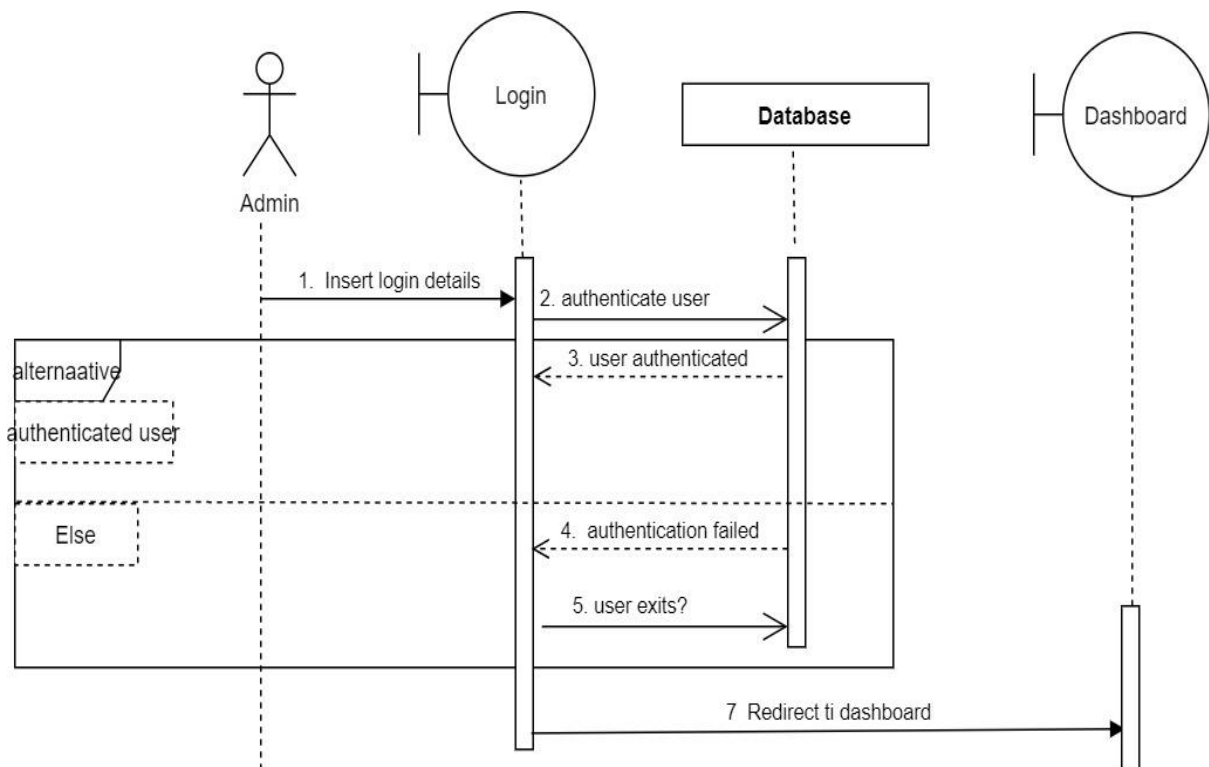
Source: Student interns

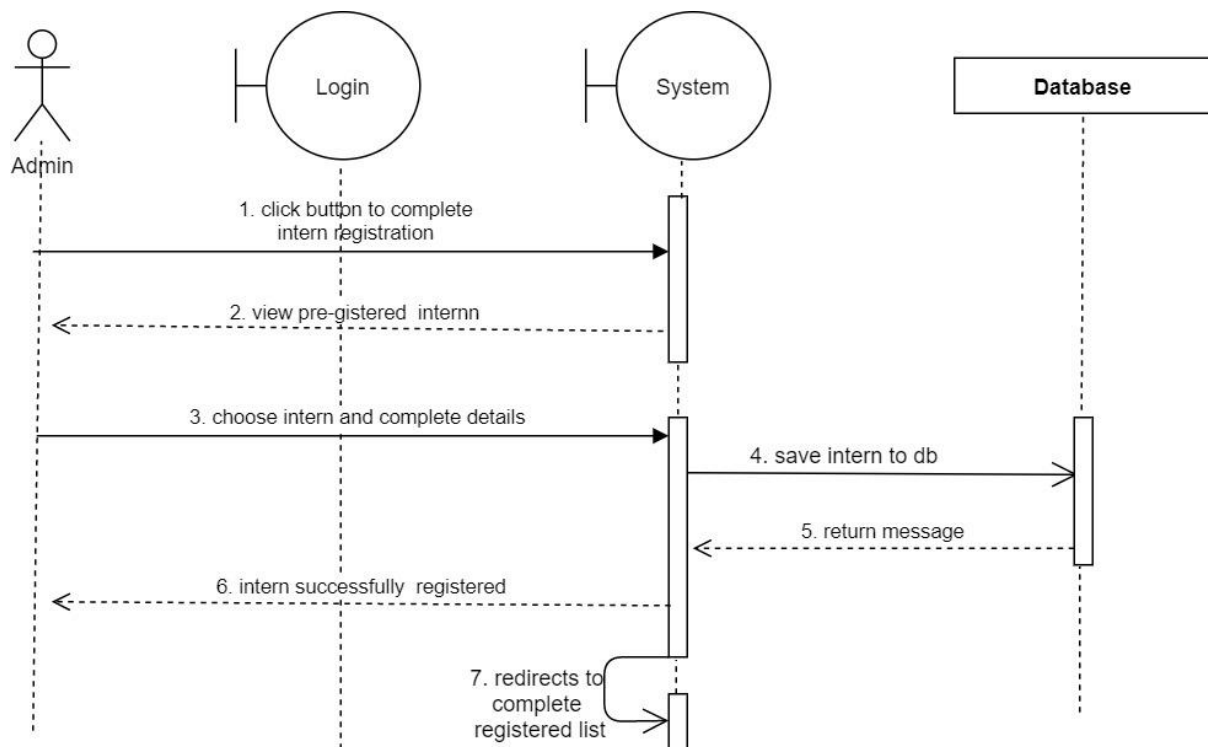
ii. Class Diagram



iii. Sequence Diagram

a). Sequence diagram for user authentication.



b). Sequence diagram to confirm intern registration.**3.1.2.3 TESTING**

The testing phase is an incredibly important part of app or website development. The problem is that the chance of a bug appearing increases with every line of code and the costs of bug fixing rise with the time. However, with a proper website testing, none of that has to happen. So, if the application testing runs smoothly, the app or website is ready to be deployed.

Web Testing Basic Steps:**Step 1: Functionality Testing**

First step of the testing phase is to make sure that the web application is functionally correct. Functionality testing checks the database connection, all links in the web pages, cookies, forms used for submitting and/or getting info from the user etc. It should be done early in the developing stages to speed up the whole app-building process and it reduces risks toward the end of the cycle. It

boils down to performing a set of tasks that can be performed both automatically with the use of a software program or manually with a human tester.

Step 2: Usability Testing

When thinking about how to test a website, the second step should be usability testing. This type of testing combines the previously mentioned functionality testing with the overall user experience. Usability testing can be done by getting external testers that simulate your expected user base, or internally by the developers themselves. If you want to perform external testing but don't know where to find suitable testers, apps like Apple's TestFlight might help you.

Step 3: Interface Testing

This web page test checks whether or not all interaction between the app server and the web server run smoothly. Not only the communication process needs to be tested, but the displaying of error messages as well. Also, this test is used to determine whether the interruptions by the server and/or by the user are handled properly.

Step 4: Compatibility Testing

In the era of portable devices and all kinds of screens, compatibility testing is important to ensure that your web app displays as it should on all device types. The developers need to run the browser compatibility test to check if the app is displayed correctly in different browsers. Another test to run is the mobile browser compatibility which, as the name suggests, is supposed to confirm that

the app displays correctly across mobile browsers. Different popular browsers include Chrome, Internet Explorer, Safari, Firefox etc.

Step 5: Performance Testing

Once you know your site is functional and responsive, you need to check its performance under a heavy load. Performance testing includes testing under different internet speeds as well as under normal and peak loads. Even in a wider context it might be a good idea to test your overall web server performance on a basic level, through a website speed automated test, as page speed influences user experience a lot, and affects your public website's SEO.

Step 6: Security Testing

Once the web application is developed, it has to be tested for security. This type of testing includes all kinds of processes to determine the app's weak points and improve them as much as possible. Normally, a series of fabricated malicious attacks are used to test how the app responds and performs under these circumstances. If security shortfalls are detected, it is important to find the best way possible to overcome them. Understanding the security goals and planning the test by considering the security needs of each of them. Identifying and listing all potential threats and vulnerabilities Preparing the threat profile based on the listed vulnerabilities and risks Preparing the test plan Automated testing on top of the manual testing Preparing the Security tests case document Carrying out

the Security test cases execution, fixing identified defects, retesting Creating a detailed report on the security testing, including the solved issues and the risks that persist.

SECTION TWO: TOOLS USED AND RESULT OBTAINED

In this section, we will talk about the reasons for the choice of tool.

PART 1: TOOLS USED

I. Visual Studio Code

- It is free and open source.
- It has an industrial level of development environment.
- It supports many other languages.
- Visual studio Code tools has a lot of plugins which are quickly updated with the latest technology that can be integrated into existing code.

II. SQLite

- It's a very light weighted database so easy to use as embedded software in devices like mobile phones, etc.
- Reading and writing operations are very fast for SQLite database. It is almost 35% faster file system.
- If you edit a small part, it only overwrites the part of the file which is changed.
- No installation required just download SQLite libraries in your computer and its ready for creating the database.
- Accessible
- Portability
- Reduce cost and Complexity.

III. Data Conception Elementary Data Dictionary

The consistency and reliability of our database (DB) is based on structuring of data that can be manipulated by the system easily. For this purpose, the database is arranged respecting normalization. The database is made up of three tables: One for collecting user data, the other for collecting data about projects worked/working on and the customer's table. The datatypes collected here are more dates, varchar, text and integer as shown in the Django Admin screenshots below.

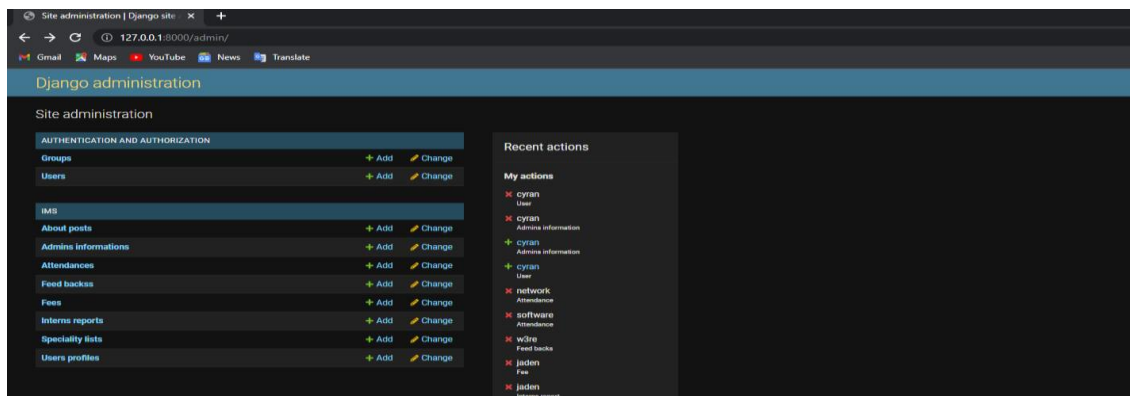


Table 2: List of all tables

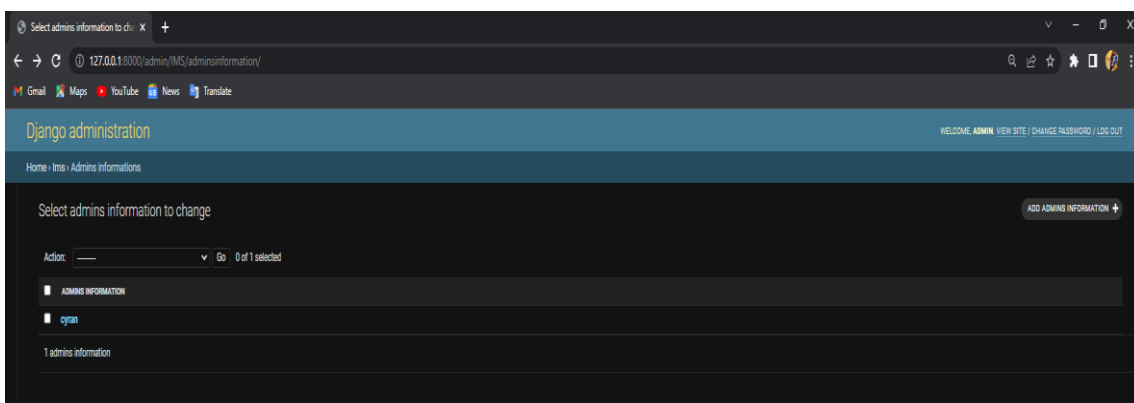


Table 3: Admins table

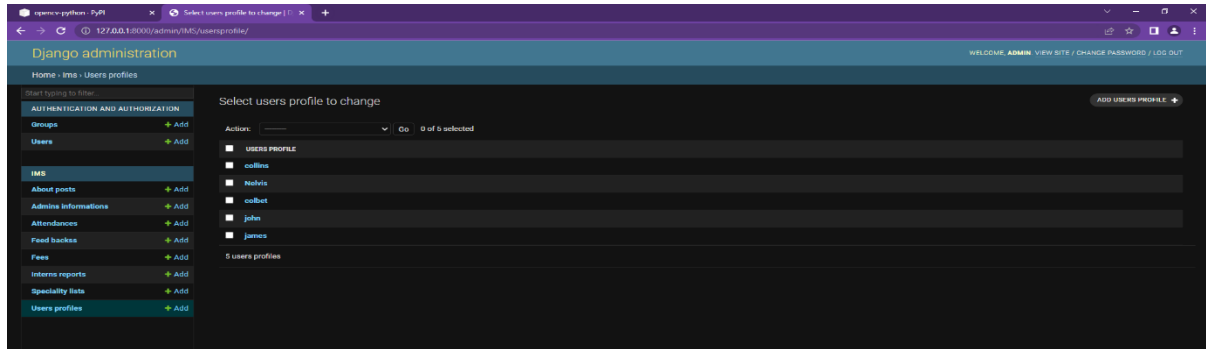


Table 4: Interns and Trainers table

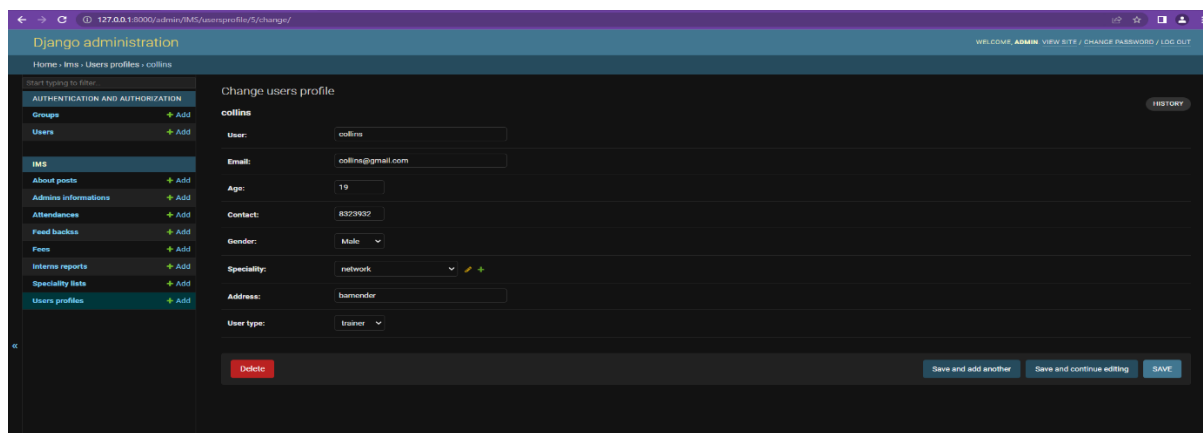


Table 5: Interns and trainers' details

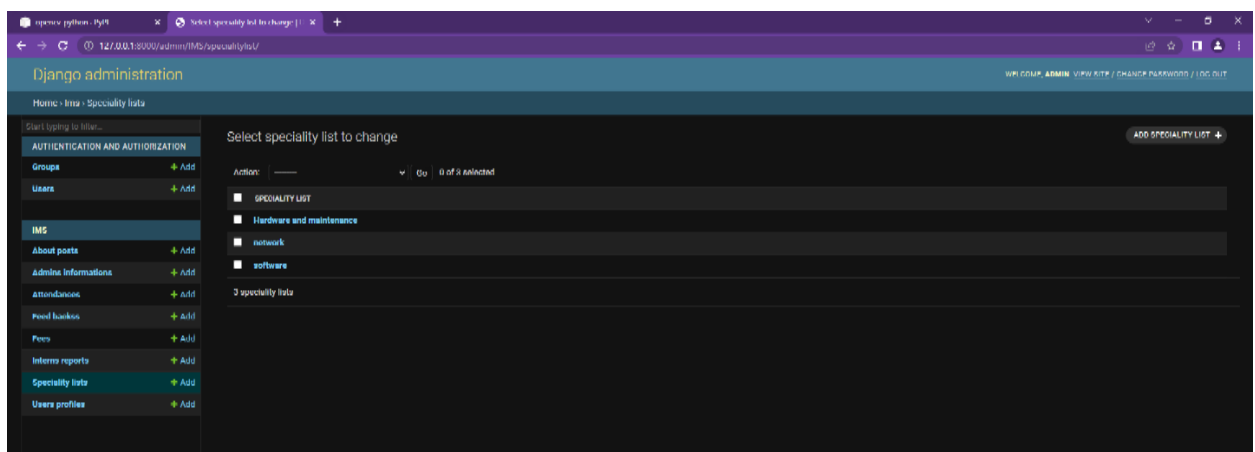


Table 6: Specialty table

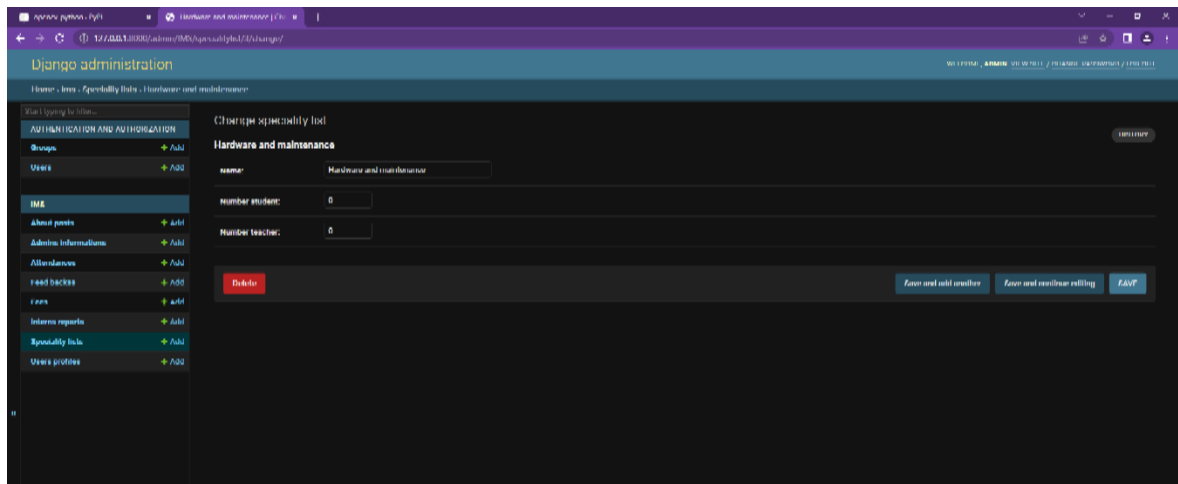


Table 7: specialty details

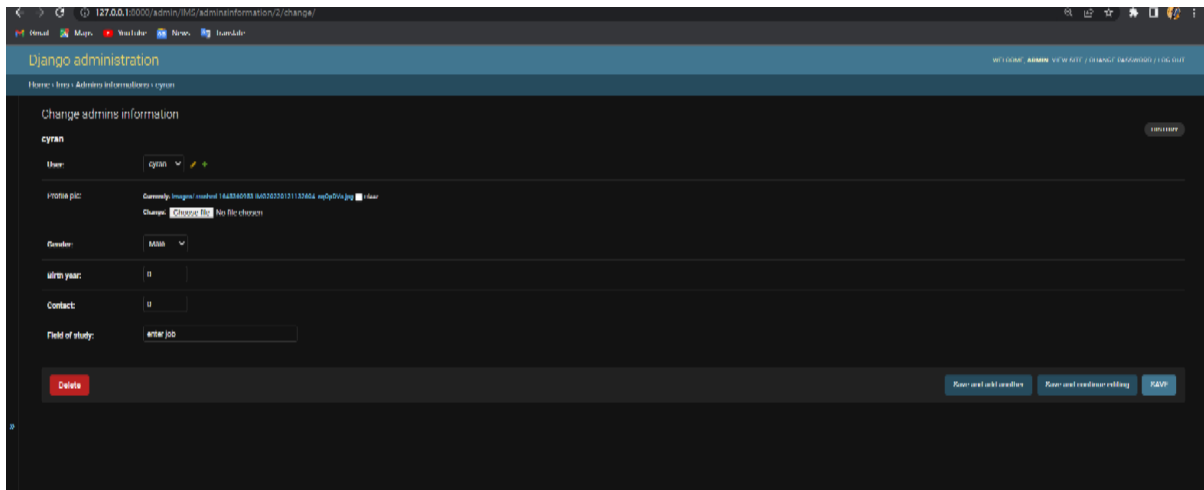


Table 8: Admins details

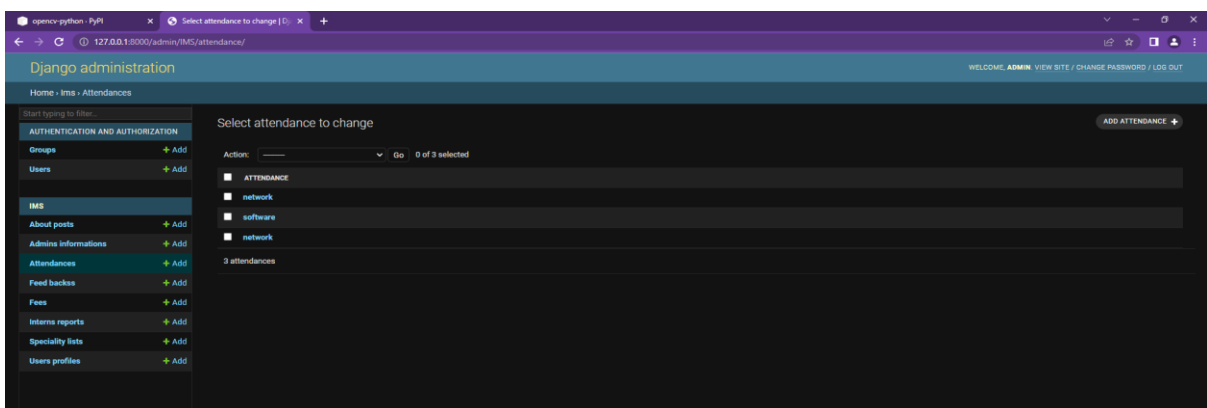
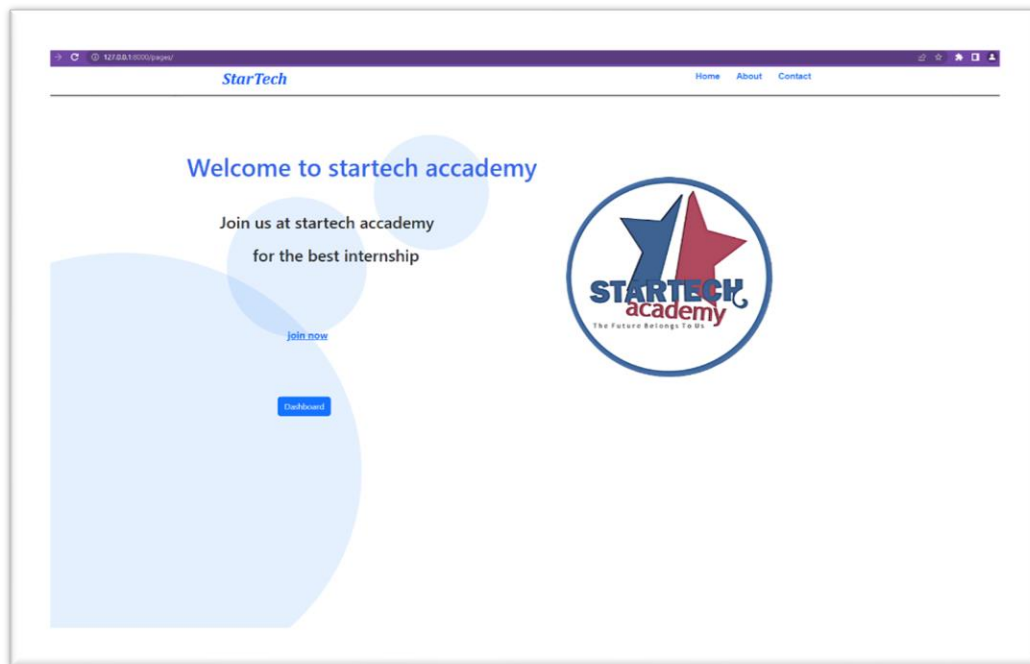
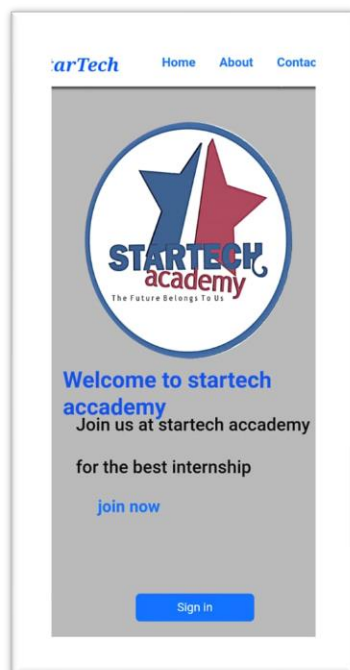


Table 8: Attendance table

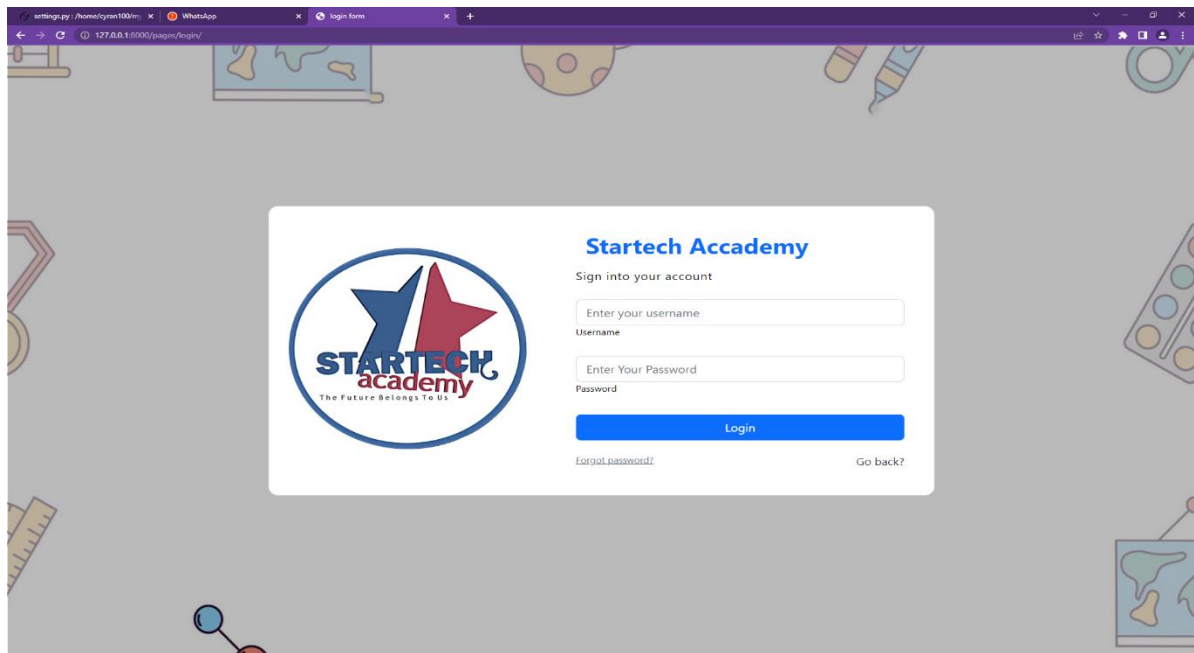
PART 2: RESULT OBTAINED



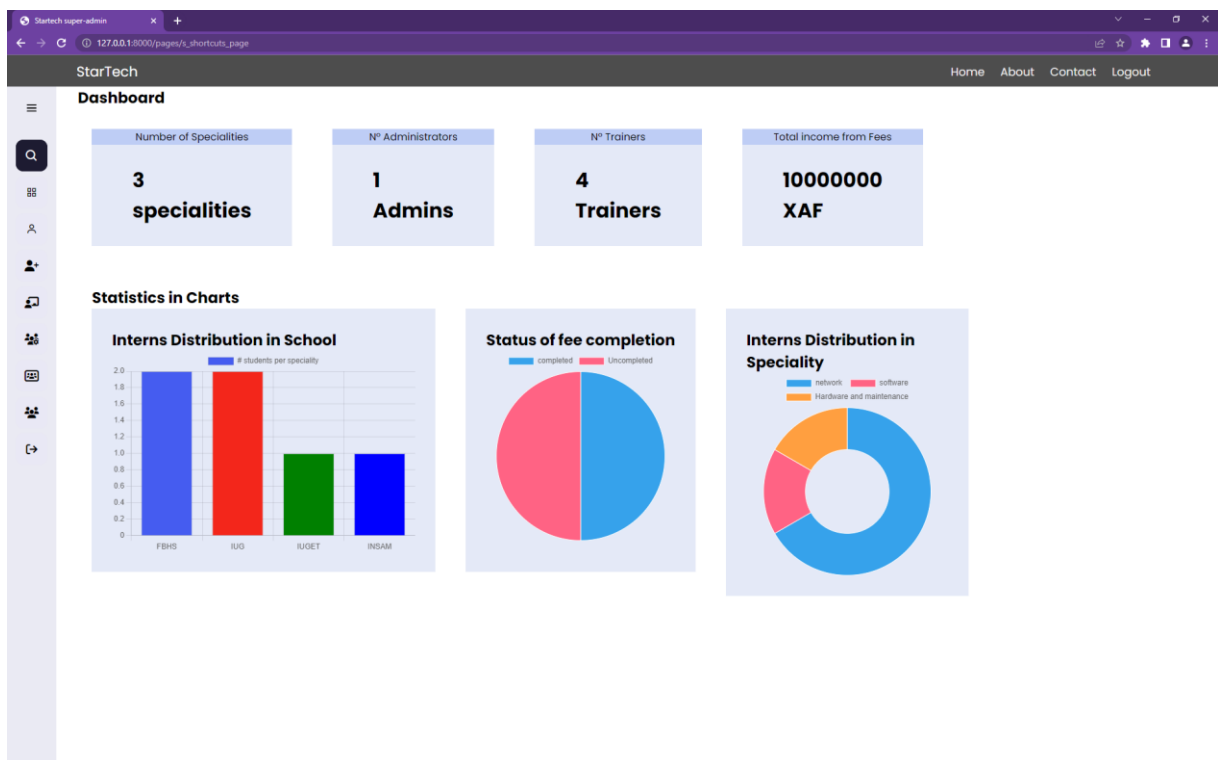
Figures 4: Welcome page



Figures 5: Responsive view of welcome page

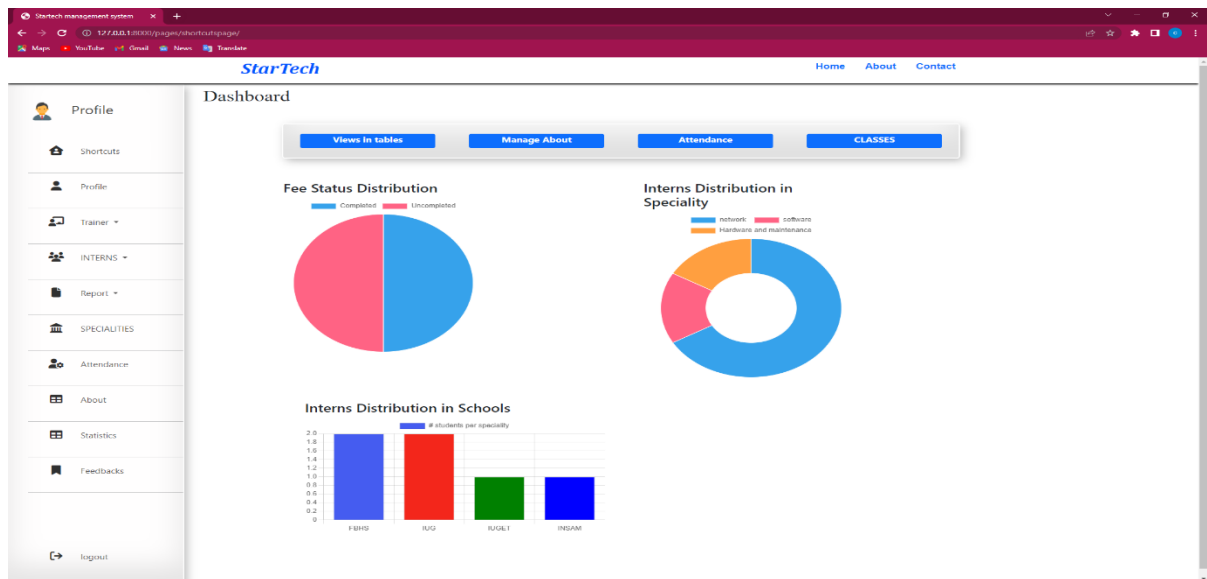


Figures 6: Login page



Figures 7: super admin Dashboard

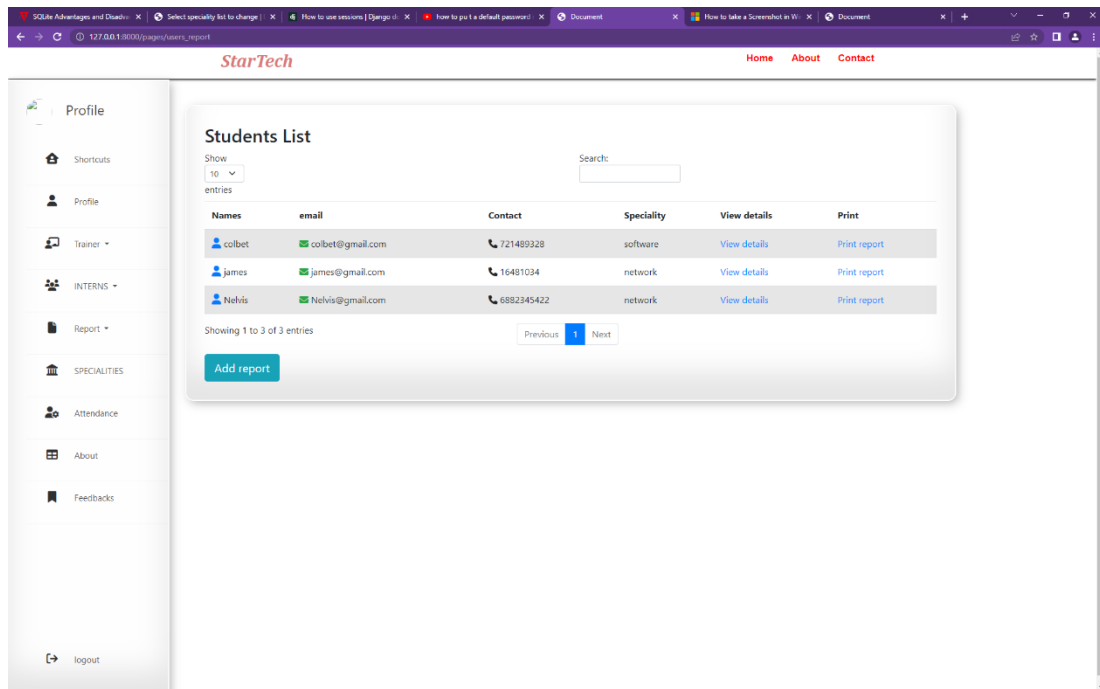
Source: Student intern



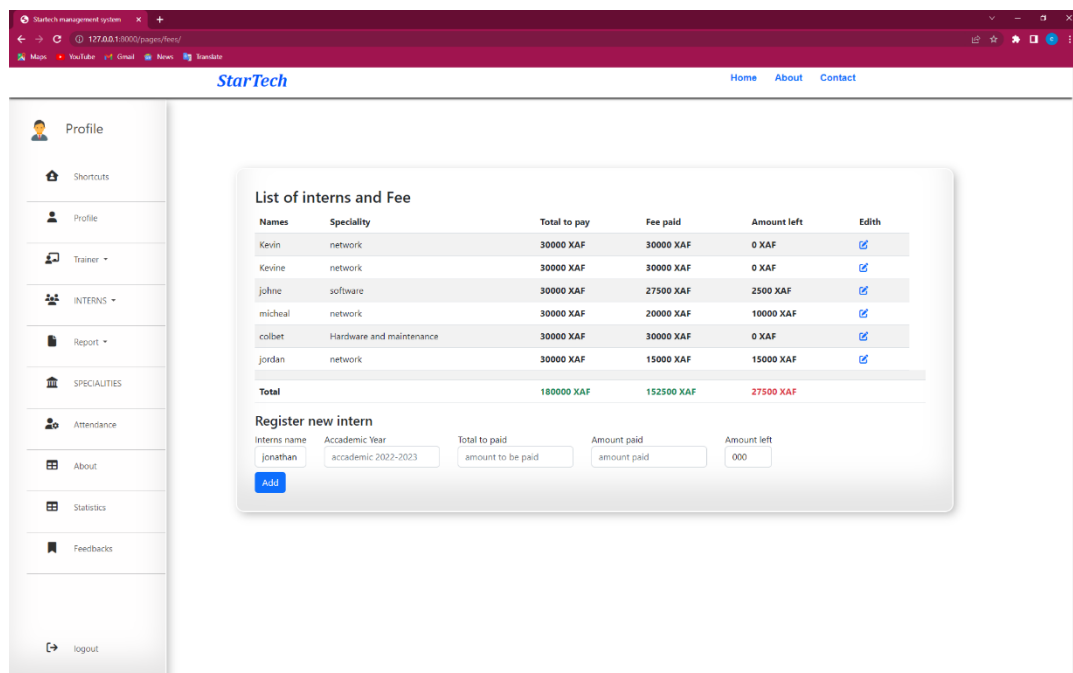
Figures 8:Admins dashboard

The screenshot shows the 'Add Administrator' form in the StarTech system. The form is titled 'Add Administrator' and includes the StarTech Academy logo. The fields to be filled are: Admin name (with a placeholder 'Admin username'), Email (with a placeholder 'email@gmail.com'), Contact (with a placeholder 'enter contact'), Field of study (with a placeholder 'enter Field'), Gender (a dropdown menu with 'Male' selected), Birthday (with a placeholder 'Birthday ex 2023-05-23'), Password (with a placeholder 'enter good password'), and Confirm Password (with a placeholder 'confirm your password'). An 'Add' button is located at the bottom of the form.

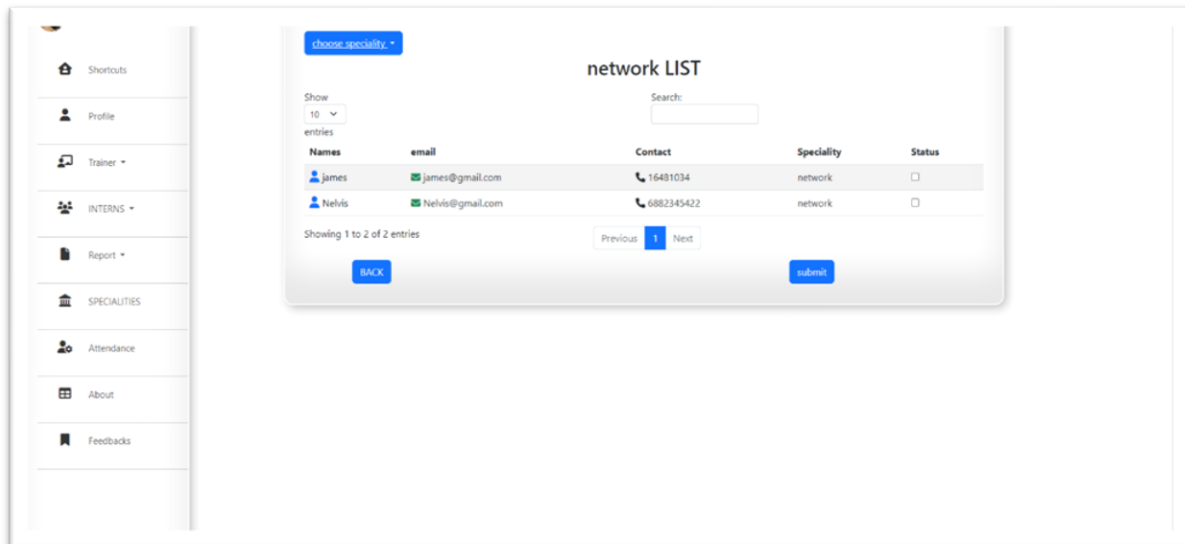
Figures 9: Add administrator.



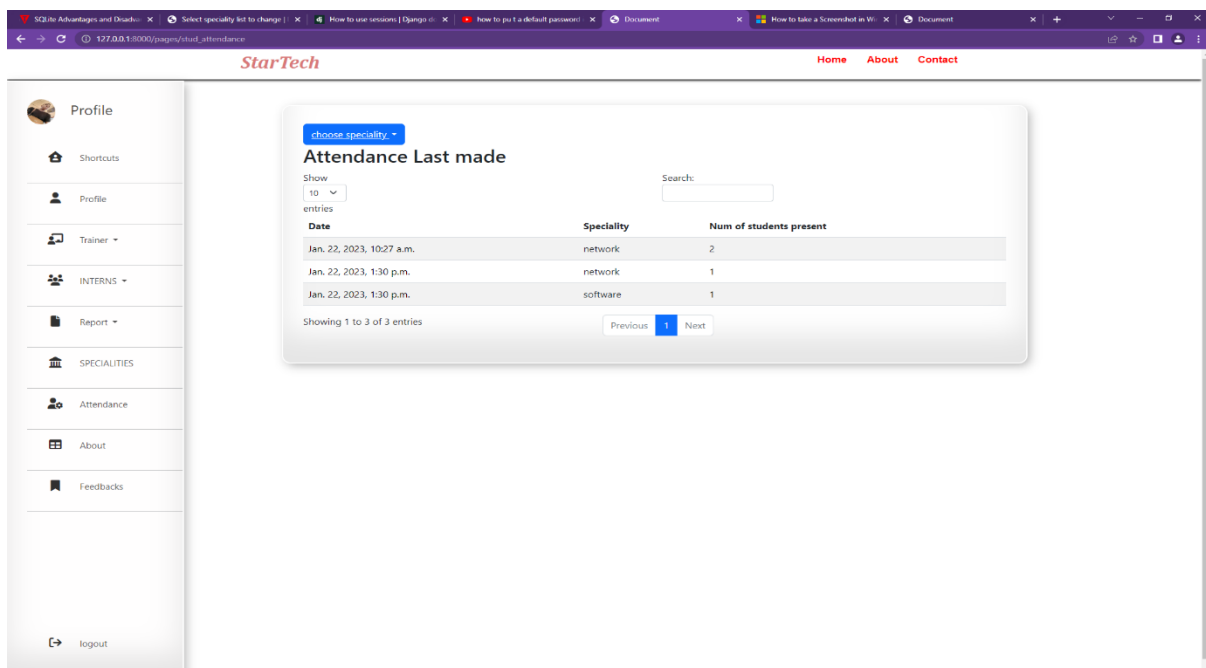
Figures 10: Created report list



Figures 11: Add interns and interns list.



Figures 12: Admin make attendance.



Figures 13: Existing attendance list

CHAPTER 4

DIFFICULTIES ENCOUNTERED AND SUGGESTIONS

This chapter is made up of two sections. Section 1 which gives the difficulties encounter on the building of the project and Section 2 which gives the suggestion.

SECTION 1: DIFFICULTIES ENCOUNTERED

- Lack of clear understanding of object-oriented concepts in Django and JavaScript.
- Problem of internet connection to access resources available online.
- Inadequate capital for transport during internship period.
- Problem of testing the application on different devices.
- Problem to install application on different operating system like MacOS X.
- Problem to add an online payment method.

SECTION 2: SUGGESTIONS

For further studies, the system will be composed of the following functionalities.

- An Online payment method.
- A mobile application for the admin to be able to access the app on his mobile phone and to avoid wastage of energy from carrying his computer anywhere.
- A comment box for the post in the about page.
- A voice assistance to easily search names of interns in the system.

GENERAL CONCLUSION

The objectives of this project were to develop an Interns Management System which will facilitates the recording of intern's information, recording daily attendance, and generating an interns report. This involve designing a user interface and a database to collectively collect and store all attendance of each specialty of the STARTECH company. This system is to replace the manual form in which the school perform attendance and the way they store information which was highly stressful and insecure. This website is divided in two main parts, the FRONT-END and the BACK-END. The front-end is made up of HTML, CSS and JAVASCRIPT, using frameworks like BOOTSTRAP and JQUERY and AJAX. The web structure and the mark-up are handled by HTML and the styling is done with CSS and BOOTSTRAP. The back end handled with pure PHP and SQL and stored in MARIADB database. Bringing this system to working was of great use to the company. My internship at STARTECH was an overall good experience. I learnt many things about the professional world though we were dealing real client's projects. As software engineers it is important to understand and then see the real world and how our software impact an organization to solve the problem of time and space complexity.

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